

Optimizing Inference Cost for Arabic–English Mathematical Reasoning in LLMs

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Abstract

Large language models (LLMs) can solve mathematical problems through prompting, yet not all reasoning is cost-justified when using paid APIs. We study prompt-only methods for overthinking control in the black-box setting across four datasets: Arabic GSM8K, English GSM8K, English SVAMP, and ArMATH, each evaluated on 100-question subsets divided into Easy, Medium, and Hard categories. We test Direct Answer, Chain-of-Thought (CoT), First Answer Anchoring (FAA-lite), Confidence-Triggered Escalation (CTE), Self-Consistency, Verification, Arabic prompt variations, and Language-Optimal Prompt Routing (LOPR). CTE achieves 90% accuracy on Arabic GSM8K at 1.14 calls/query, 96% on English GSM8K at 1.06 calls/query, and 98% on SVAMP at 1.08 calls/query. Self-Consistency leads on ArMATH at 90% but costs 2.00 calls/query. Verification consistently increases token cost without accuracy gains. Format-Controlled CoT is the best single-call Arabic method; Bilingual-Glossary CoT is consistently harmful. CTE calibration reveals that explicit low-confidence outputs are rare (<1%), so CTE succeeds mainly because high-confidence predictions are accurate, not through active uncertainty detection.

1 Introduction

Mathematical word problems involve multiple reasoning steps and are a natural challenge for LLMs, since text completion alone cannot reliably solve them. Chain-of-Thought (CoT) prompting encourages step-by-step reasoning before reaching a conclusion and substantially improves accuracy on multi-step arithmetic problems (Wei et al., 2022). Self-Consistency (SC) extends CoT by sampling multiple reasoning paths at nonzero temperature and selecting the most common final answer (Wang et al., 2022).

In black-box API use, every additional call costs money, users pay per token and per query without

access to model weights, logits, or hidden states. Our question is not whether more reasoning helps, but whether it is *worth the extra cost*. Efficient reasoning has been described as additional computation that reliably improves quality (Sui et al., 2025). Overthinking occurs when extra calls or tokens fail to produce consistent accuracy gains (Sui et al., 2025; Wei et al., 2025; Qiu and Srivastava, 2025). Most existing efficient-reasoning methods require model access or training (Wei et al., 2025; Wen et al., 2025); our methods use only text outputs and work in any black-box API.

Initial trials on GSM8K-like tasks showed that single-query CoT already yields high accuracy. Multi-sample techniques sometimes outperformed single-query CoT, but in other cases provided no improvement, echoing overthinking findings where longer or repeated reasoning yields diminishing or even detrimental accuracy gains (Sui et al., 2025; Wei et al., 2025). We operationalize overthinking as: if a technique requires additional calls or tokens without consistent accuracy improvement, it overthinks.

We evaluate prompt-only controls on four datasets spanning Arabic and English. Arabic has fewer mathematical reasoning benchmarks; ArMATH provides 6,000 Modern Standard Arabic word-problem examples (Alghamdi et al., 2022). Multilingual CoT studies show that the language of the reasoning trace affects model behavior (Shi et al., 2022), motivating separate Arabic and English analyses.

The methods we propose are intentionally minimal. FAA-lite asks the model to commit to an initial numerical answer before a brief verification step. CTE attaches a strict confidence label to the first CoT call and escalates to additional samples only when confidence is low or absent. Neither requires training, special decoding, or internal probabilities (Hu et al., 2025; Zhao et al., 2025; Ehab and Hamdi, 2026).

2 Contributions and Scope

We make six contributions, presented in order.

Black-box cost-accuracy evaluation. A black-box cost-accuracy evaluation across four datasets. Prior work usually reports only accuracy (Wei et al., 2022; Wang et al., 2022; Cobbe et al., 2021; Patel et al., 2021). We additionally report tokens per question, calls per question, confidence path counts, and per-difficulty accuracy, making the cost-accuracy tradeoff explicit. A method that improves accuracy by one point while tripling calls may be unacceptable in an API setting.

CTE calibration analysis. An empirical analysis of CTE confidence calibration separating responses by path (trusted-high, escalated-low, missing-confidence). The low-confidence path rarely fires, which has broad implications for confidence-based prompt routing. To our knowledge this is the first study reporting this calibration pattern across Arabic and English math reasoning under matched conditions.

CTE as adaptive inference rule. CTE uses one deterministic CoT-style call with a required confidence line. Trusted-high answers return immediately; low or missing confidence triggers two additional stochastic CoT samples with majority voting. CTE resembles cost-aware routing methods such as PromptWise (Hu et al., 2025) and difficulty-aware routing (Zhao et al., 2025; Ehab and Hamdi, 2026), but requires only text outputs.

FAA-lite analysis. The explicit study of FAA-lite diagnoses whether anchoring the initial answer prevents detrimental self-revisions, as suggested by past literature (Sui et al., 2025; Wei et al., 2025). FAA-lite performs better than CoT in Arabic GSM8K, but not in the other three datasets.

Arabic-specific prompt variants. We test Format-Controlled CoT, Numeral-Normalized CoT, and Bilingual-Glossary CoT only on Arabic datasets, focused on final answer extraction, mixed digit systems, and lexicon ambiguity, respectively. Only Format-Controlled CoT improves over CoT. Bilingual-Glossary CoT consistently hurts, reported as a negative result with practical implications for Arabic prompt design.

Preliminary LOPR-style analysis. LOPR-style prompting keeps the Arabic problem as input but asks the model to reason in English (Shi et al.,

2022). Results are reported as a language-routing diagnostic, not a main method. We fix $K=2$ for all final experiments and avoid dataset-specific tuning, distinguishing $K=1$ from true Self-Consistency since no voting occurs at one sample (Wang et al., 2022).

3 Related Work

CoT prompting converts direct-answer tasks into step-by-step reasoning tasks and improves accuracy on arithmetic problems in capable models (Wei et al., 2022). Self-Consistency extends CoT by sampling several reasoning paths and voting on the final answer (Wang et al., 2022). Our experiments include both, but our focus is whether extra cost is justified.

GSM8K is the standard grade-school math benchmark (Cobbe et al., 2021). SVAMP tests robustness against surface-level dataset cues (Patel et al., 2021). ArMATH is a Modern Standard Arabic word-problem dataset with 6,000 examples (Alghamdi et al., 2022). Multilingual CoT research shows that the reasoning language impacts model performance and output formatting (Shi et al., 2022), motivating our separate Arabic and English reporting.

Efficient-reasoning work includes early-exit methods (Wei et al., 2025), Budget Thinker which uses control tokens and training to impose a token budget (Wen et al., 2025), and batch prompting that amortises cost across related queries (Qiu and Srivastava, 2025), all requiring model access or training. Routing approaches such as PromptWise (Hu et al., 2025), difficulty-aware routing (Zhao et al., 2025; Ehab and Hamdi, 2026), and RADAR (Fernandez et al., 2025; Qu et al., 2026) are related but require separate routing models. CTE resembles these but uses text only.

Verbalized confidence can proxy logit-based estimates (Kadavath et al., 2022; Tian et al., 2023; Lin et al., 2022); our calibration analysis extends this to Arabic and English math tasks without any fine-tuning. Prompt engineering reproducibility (White et al., 2023) informed our logging and versioning protocol.

4 Methods

All methods share the same model, output format (#### <number>), and strict parser (Cobbe et al., 2021). Methods fall into three cost tiers: one-call (Direct, CoT, FAA-lite, Arabic variants,

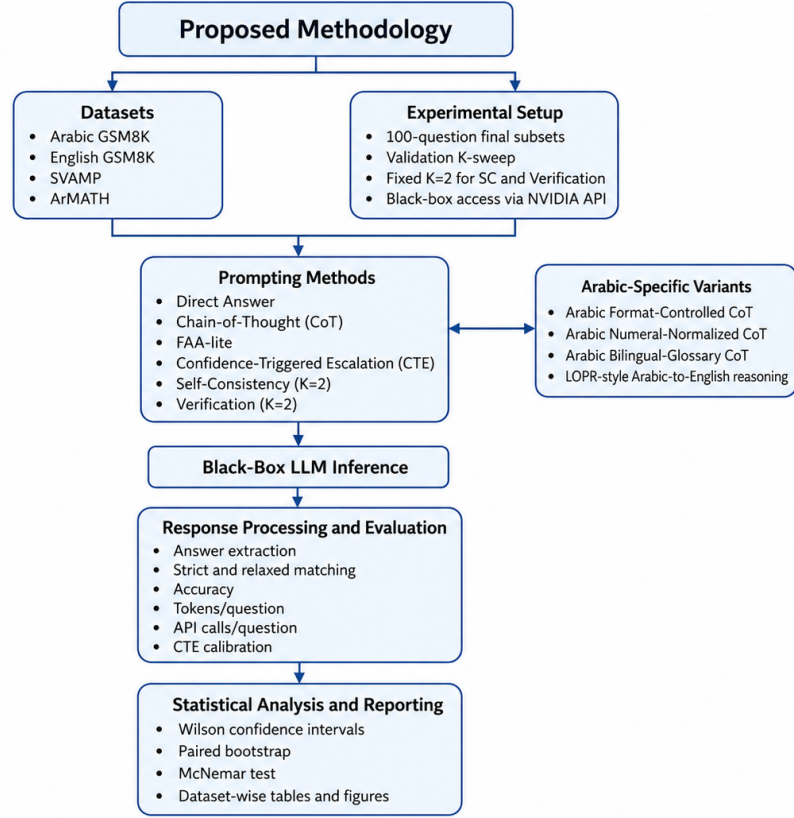


Figure 1: Overview of the experimental pipeline, from dataset selection and prompting methods through black-box LLM inference to statistical evaluation.

LOPR), adaptive up to 3 calls (CTE), and fixed multi-call (SC $K=2$, Verification). Figure 1 provides a high-level overview of the full experimental pipeline, from dataset selection through statistical evaluation.

Direct Answer. Final number only; underthinking baseline (Sui et al., 2025).

Chain-of-Thought (CoT). Step-by-step reasoning; primary single-call baseline (Wei et al., 2022). All proposed methods are compared against CoT.

Self-Consistency $K=2$. Two independent CoT samples at $T=0.7$, majority vote (Wang et al., 2022). $K=2$ was chosen via a held-out validation sweep (Section 5).

Verification $K=2$. Two candidates generated, then a separate verifier selects the final answer. Total cost ≈ 3 calls per question.

FAA-lite. Single call: model commits to an initial numerical answer, then briefly verifies and produces the final answer line. Tests whether anchoring reduces harmful self-revision.

CTE. Confidence-Triggered Escalation. A CoT call requiring an explicit `CONFIDENCE: High` or `CONFIDENCE: Low` line. Trusted-high answers return immediately; low or missing confidence triggers two additional stochastic CoT samples with majority voting. Maximum 3 calls; average approximately 1.06–1.16 calls depending on the dataset. Confidence detection is strict: Only an exact `CONFIDENCE: line` is accepted.

Arabic variants (Arabic datasets only). *Format-Controlled CoT*: enforces Western-Arabic numerals and a single final answer line. *Numeral-Normalized CoT*: converts Eastern Arabic numerals. *Bilingual-Glossary CoT*: prepends Arabic mathematical term definitions.

LOPR-style. Language-Optimal Prompt Routing. The Arabic problem is kept as input, but the model is instructed to reason in English and return the answer in the required format. Unlike full routing systems that select the reasoning language per-problem based on a difficulty signal, this condition always routes to English and serves as a diagnostic of cross-lingual reasoning potential rather than a

deployable routing strategy. Reported as a preliminary analysis from an earlier experimental run, not as a method in the main evaluation tables.

The eight core methods organize into three cost tiers. Direct Answer, CoT, FAA-lite, Format-Controlled CoT, Numeral-Normalized CoT, Glossary CoT, and LOPR-style are one-call methods. CTE is adaptive with a maximum of three calls. Self-Consistency and Verification are fixed multi-call methods at two and three calls, respectively.

Both English and Arabic prompts follow the same structure. Full prompt templates for all remaining methods (Direct Answer, FAA-lite, Verification, Numeral-Normalized CoT, and Bilingual-Glossary CoT), along with the complete experimental code, are publicly available at [URL](#)

5 Experimental Setup

Model & API. All experiments use `openai/gpt-oss-20b` accessed via the NVIDIA API. Deterministic first-pass calls (Direct, CoT, FAA-lite, the CTE first pass, and the verifier) use $T=0$. Stochastic samples (Self-Consistency, Verification, and CTE escalation) use $T=0.7$, $\text{top-}p=1.0$. Rate limit: 30 requests per minute. These parameters are reported because inference cost is part of the object of study.

Datasets. Arabic GSM8K (Omarificial-Intelligence-Space, 2025), English GSM8K (Cobbe et al., 2021), SVAMP (Patel et al., 2021), and ArMATH (Alghamdi et al., 2022). Each dataset uses a 100-question final test set stratified into Easy (34 questions), Medium (33), and Hard (33). Difficulty is derived from reasoning-step count for GSM8K-style datasets, equation structure for SVAMP, and equation-template tertiles for ArMATH (Alghamdi et al., 2022). Difficulty labels are used only for stratified sampling and per-difficulty reporting.

K selection. A preliminary K -sweep was conducted once on a held-out 50-question Arabic validation subset with $K \in \{1, 2, 4, 8, 16\}$. $K=4$ gained 4 accuracy points over $K=2$ (92% vs. 88%) but nearly doubled token cost; $K=8$ and $K=16$ returned to 90% with no stable improvement. We fix $K=2$ as the cost-efficiency elbow to avoid dataset-specific tuning. The $K=2$ choice is a deployment-oriented compromise: $K=4$ might give 2–4 additional accuracy points at twice the cost, but picking the elbow of the validation curve preserves compa-

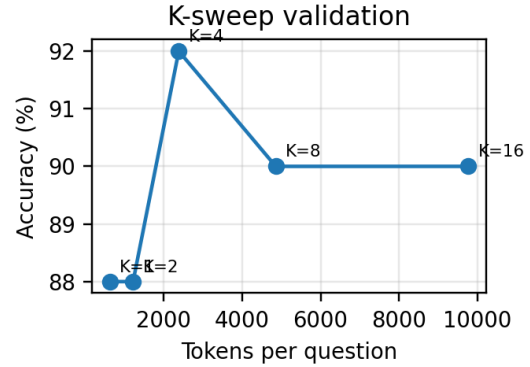


Figure 2: Plots accuracy against token cost K -sweep on different values of ($K=1,2,4,8,16$)

Table 1: K -sweep summary ($n=50$ Arabic GSM8K validation set).

K	Acc. %	Tok./Q	Calls/Q	Note
1	88.0	634	1.00	single stochastic ref.
2	88.0	1223	2.00	min. true multi-sample
4	92.0	2387	4.00	higher acc., high cost
8	90.0	4857	8.00	no stable gain
16	90.0	9762	16.00	large cost, no gain

rability across datasets. A dedicated $K=4$ sensitivity check is left for future work. Figure 2 plots accuracy against token cost.

LOPR-style condition handling. The LOPR-style condition uses Arabic problems and is part of the Arabic analysis, not the English benchmarks. LOPR-style numbers come from a test run with a separate experimental pipeline and are reported in Section 7.1 rather than merged into the main Arabic tables.

Logging and reproducibility. Raw model responses, fallback-extraction cases, API errors, prompt versions, cache keys, and per-problem records were saved for every run. Prompt-only experiments are difficult to reproduce when prompts or parsers change (White et al., 2023).

Metrics and Statistics. We report accuracy, tokens per question, calls per question, and fallback-extraction rate. Where two methods produce similar accuracy, we additionally report paired bootstrap differences and McNemar exact p -values from per-problem records. A two-point gap on 100 problems is not strong evidence without a paired test.

We report accuracy, tokens per question, calls per question, and fallback-extraction rate. Where two methods produce similar accuracy, we addi-

Table 2: Arabic GSM8K ($n=100$). **Bold** = best accuracy.

Method	Acc. %	Tok./Q	Calls/Q
Direct Answer	65.0	344	1.00
CoT	84.0	701	1.00
FAA-lite	87.0	707	1.00
CTE	90.0	893	1.14
Self-Consistency $K=2$	88.0	1368	2.00
Verification $K=2$	86.0	1768	2.98
Format-Controlled CoT	87.0	533	1.00
Numeral-Normalized CoT	86.0	765	1.00
Bilingual-Glossary CoT	72.0	865	1.00

Table 3: English GSM8K ($n=100$).

Method	Acc. %	Tok./Q	Calls/Q
Direct Answer	85.0	302	1.00
CoT	95.0	488	1.00
FAA-lite	93.0	550	1.00
CTE	96.0	583	1.06
SC $K=2$	97.0	1007	2.00
Verification $K=2$	94.0	1392	2.98

tionally report paired bootstrap differences and McNemar exact p-values from per-problem records. A two-point gap on 100 problems is not strong evidence without a paired test. Unless otherwise noted, bracketed values in the results denote 95% Wilson confidence intervals

6 Results

6.1 Arabic GSM8K

CTE achieves the highest accuracy at 90% [0.83, 0.94] with only 1.14 calls/question (Table 2). CoT reaches 84% [0.76, 0.90] at one call. Format-Controlled CoT reaches 87% at 533 tokens, below CoT’s 701, making it the most token-efficient single-call method. Bilingual-Glossary CoT is harmful at 72% [0.63, 0.80] (McNemar $p=0.008$).

6.2 English GSM8K

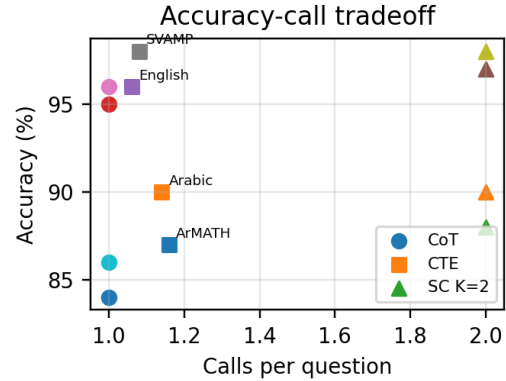
CoT is already strong at 95% [0.89, 0.98] (Table 3). SC reaches 97% [0.92, 0.99] but requires 2.00 calls; CTE matches 96% [0.90, 0.98] in 1.06 calls. No method significantly beats CoT (Direct Answer is significantly worse $p=0.006$). CTE’s value here is cost reduction.

6.3 SVAMP

CTE and SC both reach 98% [0.93, 0.99] (Table 4); CTE uses 1.08 calls vs. SC’s 2.00 and 488 tokens vs. SC’s 833. Verification reaches 95% at 3.00 calls, the clearest demonstration that extra calls do not guarantee accuracy gains.

Table 4: English SVAMP ($n=100$).

Method	Acc. %	Tok./Q	Calls/Q
Direct Answer	91.0	233	1.00
CoT	96.0	433	1.00
FAA-lite	94.0	425	1.00
CTE	98.0	488	1.08
SC $K=2$	98.0	833	2.00
Verification $K=2$	95.0	1117	3.00

**Figure 3:** Accuracy vs. calls/question for CoT, CTE, and SC $K=2$ across all four datasets. CTE matches SC at near single-call cost on three of four datasets.

6.4 ArMATH

SC achieves the best accuracy at 90% [0.83, 0.94] (Table 5); the bootstrap CI vs. CoT excludes zero ([+0.01, +0.08]), the only significant gain on this dataset. Format-Controlled CoT is the best single-call method at 88% [0.80, 0.93] with 477 tokens. Bilingual-Glossary CoT is again harmful (bootstrap -0.06 , CI $[-0.12, -0.01]$). CoT drops from 94.1% on Easy to 72.7% on Hard (Table 6).

Figure 4 compares Arabic prompt variants across both Arabic datasets; Figure 3 shows accuracy vs. calls across all four datasets.

7 Calibration Analysis

Table 7 breaks CTE responses into three paths. Across 400 total runs, only 2 responses produced `CONFIDENCE: Low`. Escalation fires almost entirely on *missing-confidence* cases where the model failed to follow the format, and those cases have very low accuracy (12.5–66.7%). Trusted-high cases are accurate 93.5–99.0% of the time.

Three observations follow:

First, trusted-high cases are highly accurate: 93.55% on Arabic GSM8K, 97.94% on English GSM8K, 98.96% on SVAMP, and 93.48% on ArMATH. When the model claims high confidence on its own initiative, it is usually right.

Table 5: ArMATH ($n=100$).

Method	Acc. %	Tok./Q	Calls/Q
Direct Answer	82.0	262	1.00
CoT	86.0	559	1.00
FAA-lite	84.0	509	1.00
CTE	87.0	735	1.16
SC $K=2$	90.0	1194	2.00
Verification $K=2$	87.0	1503	2.92
Format-Controlled CoT	88.0	477	1.00
Numeral-Normalized CoT	86.0	660	1.00
Bilingual-Glossary CoT	80.0	732	1.00

Table 6: ArMATH accuracy by difficulty level.

Method	Easy	Med.	Hard
CoT	94.1	90.9	72.7
Format-Ctrl CoT	97.1	90.9	75.8
CTE	97.1	87.9	75.8
SC $K=2$	97.1	93.9	78.8

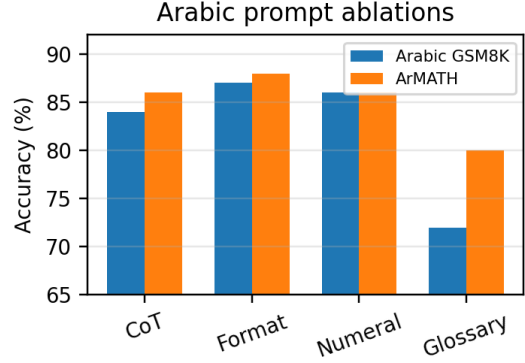
Second, missing-confidence cases have very low accuracy: 33.33% on Arabic GSM8K, 33.33% on English GSM8K, 66.67% on SVAMP, and 12.50% on ArMATH. Escalation does not recover these problems. The two extra stochastic samples do not reliably converge to the right answer because the model is already confused.

Third, the explicit low-confidence path is essentially empty. Across all four datasets, we observed exactly two `CONFIDENCE: Low` cases out of 400 total CTE runs. Both were answered correctly after escalation, but $n=2$ is not enough to draw any conclusion. These three confidence paths are visualized in Figure 5.

The practical implication is that CTE works for a reason different from its design. It was intended to detect uncertain responses and spend extra compute on them. In practice, the model rarely produces uncertain responses on its own, and when it fails to follow the output format the extra compute does not help. CTE delivers cost savings because the trusted-high accuracy is high, not because the gating mechanism actively recovers wrong answers (Kadavath et al., 2022; Tian et al., 2023; Lin et al., 2022). Without explicit calibration training, the model defaults to an overconfident high-confidence stance on grade-school math, with low-confidence emission too rare to be useful as a gating signal.

7.1 LOPR-style Preliminary Analysis

We conducted a prior LOPR-style experiment on Arabic GSM8K. The accuracy was 0.84 [0.76, 0.90] at 661 tokens per question and one

**Figure 4:** Accuracy of Arabic-specific prompt variants on Arabic GSM8K and ArMATH. Format-Controlled CoT achieves higher accuracy than standard CoT. Bilingual-Glossary CoT is the lowest-performing variant on both datasets.**Table 7:** CTE confidence-path calibration.

Dataset	Path	N	Acc %	Calls
Ar.GSM8K	trusted_high	93	93.6	1.00
Ar.GSM8K	escalated_low	1	100.0	3.00
Ar.GSM8K	missing_conf.	6	33.3	3.00
En.GSM8K	trusted_high	97	97.9	1.00
En.GSM8K	missing_conf.	3	33.3	3.00
SVAMP	trusted_high	96	99.0	1.00
SVAMP	escalated_low	1	100.0	3.00
SVAMP	missing_conf.	3	66.7	3.00
ArMATH	trusted_high	92	93.5	1.00
ArMATH	missing_conf.	8	12.5	3.00

call. Accuracy by difficulty: 0.91 on Easy, 0.82 on Medium, and 0.79 on Hard. The Easy and Medium scores match those from the Arabic CoT (Easy = 0.91, Medium = 0.85).

We report LOPR-style as a diagnostic only. The qualitative takeaway is that routing the reasoning language to English does not preserve accuracy on the hardest Arabic problems. A rigorous evaluation requires paired Arabic-English problems run through the identical pipeline, which is left for future work.

8 Discussion

CTE is useful when CoT has headroom and high-confidence answers are reliable. Arabic GSM8K satisfies both conditions: CoT sits at 0.84, and trusted-high accuracy is 0.94. CTE produces a +0.06 gain at 1.14 calls. On SVAMP, the headroom is small, but the high-confidence accuracy is even higher (0.99), so CTE matches Self-Consistency at near single-call cost. On English GSM8K, the headroom is minimal (CoT 0.95), and CTE delivers cost reduction rather than accuracy gain. On ArMATH, the trusted-high accuracy is 0.93, but

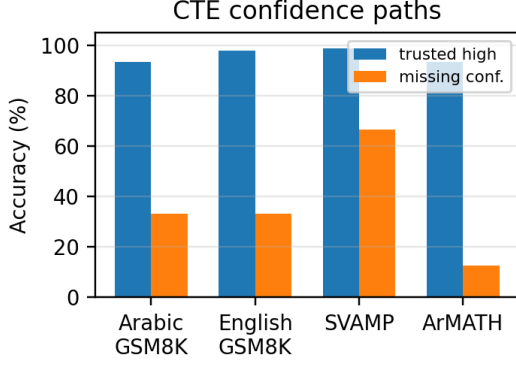


Figure 5: CTE confidence-path calibration. Trusted high-confidence answers are much more reliable than missing-confidence cases across all four datasets.

the model is still wrong often enough that Self-Consistency’s exploration of alternative reasoning paths pays off.

Verification is consistently not cost-effective. It costs about three calls per question and does not produce the optimal accuracy on any of the four datasets. It performs worse than CoT on English GSM8K. On ArMATH, it matches CTE at 0.87 but requires 2.5 times the number of calls needed by CTE.

FAA-lite is a targeted intervention, not a general improvement. It improves performance on Arabic GSM8K (+0.03 relative to CoT, bootstrap CI [0.01, 0.08]) but not on the other three. This implies that errors on Arabic GSM8K may be due to answer flipping in the case of Arabic GSM8K. The errors in answering on other datasets are not in the verifying and revising phase.

Arabic prompt design matters more than expected. Format-Controlled CoT outperforms other methods on both Arabic datasets in terms of precision as well as the number of tokens, without any additional API calls. Bilingual-Glossary CoT performs considerably worse on both datasets. The gloss introduces additional instruction weight, which causes distraction from mathematics.

$K=2$ versus $K=4$. Table 1 shows that $K=4$ attains a validation accuracy of 92%, compared to 88% for $K=2$. $K=4$ requires twice the token cost for an increase of four points on a 50-question validation set, within the bounds of the bootstrapping noise band.

Toolkit framing. Our results do not support a single best prompt-only method. They support a toolkit:

- If only one call is allowed on Arabic, use Format-Controlled CoT.
- If the model follows confidence formatting and the application tolerates occasional escalation, use CTE.
- If maximum accuracy is more important than cost and the dataset is hard (such as ArMATH), Self-Consistency $K=2$ is justified.
- For Arabic systems with cross-lingual potential, LOPR-style prompting can be used as a diagnostic but not yet as a deployment method.

9 Error Analysis

Failure logs reveal four recurring categories.

Arithmetic errors. Correct reasoning structure but wrong computation; present in both languages and not addressable through formatting.

Operation-selection errors. Wrong arithmetic relationship between variables; a core failure mode in word problems (Patel et al., 2021).

Extraction errors (Arabic). Eastern Arabic numerals, spelled-out numbers, or extra numerals after the #### marker. Format-Controlled CoT eliminates most of these, explaining why it saves tokens while improving accuracy.

Confidence-format failures in CTE. Missing `CONFIDENCE:` lines route to escalation, but escalated accuracy remains low (Section 7). These are not low-confidence outputs, the model generated a correct reasoning chain but failed to include the required format line. The strict parser correctly refuses to trust these responses and sends them to escalation, but the two extra stochastic samples do not reliably converge to the correct answer when the model is already confused.

Verification errors. The model generates a plausible candidate but the separate verification stage selects the wrong final number. These cases are expensive because the user pays for both stages while the final answer remains wrong. They partly motivated the FAA and CTE designs, although neither fully eliminates them.

10 Limitations

One model. All experiments use `gpt-oss-20b`. Results reflect behavior on

that model under the NVIDIA API and are not a universal property of all LLMs. Different model families differ in confidence calibration and multilingual reasoning (Shi et al., 2022; Sui et al., 2025). Replicating the protocol on additional models is important future work.

100-question subsets. Final test sets contain 100 examples per dataset. This is enough for cost-accuracy patterns but small for fine-grained accuracy comparisons. We interpret close differences conservatively and rely on paired bootstrap and McNemar tests for significance claims.

Unpaired Arabic and English. The Arabic and English problem sets are not aligned item by item, which is why we report them separately. A paired design would require verified Arabic-English problem pairs. Machine-translating SVAMP into Arabic was rejected because translation quality would become a confound. Using ArMATH avoids that problem at the cost of cross-dataset comparison.

LOPR-style not on the same pipeline. The LOPR-style condition is reported from a test run, not the main 4-dataset evaluation. A reviewer-grade LOPR-style result requires the same pipeline with paired Arabic-English problems.

Binary confidence labels. CTE uses a two-level confidence label. The calibration analysis shows that low-confidence outputs are very rare. A graded numeric scale or a separate lightweight uncertainty prompt might fire the escalation path more usefully. Such extensions remain in the black-box setting if they use only text outputs.

FAA-lite is a one-call approximation. A full FAA design would use a separate commitment call followed by an anchored verification call. We used the one-call version to keep the call cost equal to CoT, which limits the comparison but matches the efficiency theme of the paper.

Benchmark contamination. Contamination cannot be ruled out for public benchmarks such as GSM8K and SVAMP (Cobbe et al., 2021; Patel et al., 2021). We therefore focus on relative method behavior under matched conditions rather than absolute model capability.

11 Conclusion

Prompt-only overthinking control is effective but dataset-dependent. CTE matches or exceeds Self-

Consistency at substantially lower cost on Arabic GSM8K, English GSM8K, and SVAMP. Self-Consistency $K=2$ remains the strongest method on ArMATH. Arabic Format-Controlled CoT is the best single-call method on both Arabic datasets; Bilingual-Glossary CoT consistently hurts and should be avoided. Calibration analysis reveals that CTE succeeds mainly because the model’s unprompted high-confidence claims are accurate, not because the gating mechanism actively detects uncertainty.

The practical recommendation is a toolkit, not a single prompt: Format-Controlled CoT for single-call Arabic deployment; CTE when the model follows confidence formatting and occasional escalation is acceptable; Self-Consistency $K=2$ when maximum accuracy on difficult Arabic problems justifies the additional cost. This finding warrants future research into more sophisticated confidence-scoring schemes, paired Arabic-English datasets, and replication across other model families to address the calibration limitations identified here.

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